

Large Language Models in Academic Research: A Review of Collaboration, Barriers, Researcher Archetypes, and Opportunities for Innovation

Prepared for James Barnes

July 2026

Abstract

Between the release of ChatGPT in late 2022 and mid-2026, large language models (LLMs) moved from novelty to infrastructure in academic research. This review synthesizes roughly seventy empirical, bibliometric, and position sources across four questions: how researchers use LLMs to *collaborate*, what *barriers* constrain that use, which *archetypes* of researcher emerge, and where the *opportunities for innovation* lie. The evidence shows adoption that is broad but shallow and uneven: population-scale text analyses put LLM-assisted writing at roughly one in seven biomedical abstracts and up to a fifth of computer-science abstracts by 2024, while multi-wave surveys record tool usage rising from about 57% to 84% of researchers in a single year. That usage is dominated by consumer chat interfaces rather than IDE-integrated coding assistants or autonomous agents, and the productivity evidence is openly contradictory—a 55% speedup in one controlled coding trial against a 19% *slowdown* for experienced developers in another. Yet the same corpus documents fabricated citations, a fragmented and often prohibitive policy landscape, sharp demographic and career-stage gradients in who adopts, and a live debate over whether LLMs homogenize ideas and erode understanding even as they raise individual output. Against this backdrop we evaluate a specific hypothesis: that the future of science resembles “a big group chat of the smartest minds glued together by frontier models.” We find the literature offers strong support for the *gluing* mechanism—LLMs already act as connective tissue across the research workflow and, in multi-agent form, as collaborators that generate wet-lab-validated discoveries—but a serious empirical challenge to the *group chat* image: the best science-of-science evidence to date finds AI-augmented researchers publish and cite more while engaging *fewer* human collaborators and narrowing the collective frontier of topics. The metaphor is directionally right about connectivity and wrong, so far, about sociality.

Contents

1	Introduction	2
1.1	Motivation and guiding hypothesis	2
1.2	Scope and method	3
1.3	A note on terminology	3
2	Adoption: Who Is Using LLMs, and for What	3
2.1	Survey evidence	3
2.2	Bibliometric forensics	4
2.3	The shape of usage	4

3	The Interaction Surface: Subscriptions, Assistants, and Agents	4
3.1	Consumer chat leads; IDE-integrated tools trail	4
3.2	The productivity contradiction	5
3.3	The autonomy spectrum: assistant to agent	5
3.4	Access economics and the freemium divide	6
3.5	Implication for the group-chat hypothesis	6
4	LLMs in Research Collaboration	6
4.1	Human–AI teaming in scientific work	6
4.2	Multi-agent and autonomous “AI scientist” systems	7
4.3	LLMs and the structure of scientific teams	8
5	Barriers to Use	8
5.1	Accuracy, hallucination, and trust	8
5.2	Policy, authorship, and research integrity	9
5.3	Equity, access, and skills	9
5.4	Epistemic risks	9
6	Researcher Archetypes	10
6.1	Demographic and disciplinary segmentation	10
6.2	Behavioral typologies: centaurs, cyborgs, and self-automators	10
6.3	Attitudinal archetypes and the trust–use gap	11
7	Opportunities for Innovation	11
7.1	Ideation and hypothesis generation	11
7.2	Literature synthesis and search	11
7.3	Writing, coding, and analysis	12
7.4	Peer review and evaluation	12
7.5	Autonomous discovery	12
8	Discussion: Is the Future of Science a Group Chat?	12
8.1	The “glue” claim is well supported	12
8.2	The “group chat” claim faces a direct empirical challenge	13
8.3	Reconciling the two	13
8.4	What would confirm the hypothesis	13
9	Limitations of This Review	13
10	Conclusion	14

1 Introduction

1.1 Motivation and guiding hypothesis

Large language models entered scientific practice faster than almost any prior research technology. Within three months of ChatGPT’s public release, a questionnaire of Nature readers found that roughly 80% had already tried such a tool and more than a fifth were using one regularly (Owens, 2023). Within two years, population-scale analyses of the published record could measure the fingerprint of LLM-assisted writing directly in millions of abstracts (Liang et al., 2025; Kobak

et al., 2025). The question is no longer *whether* academics use LLMs but *how*, *who* does so, *what holds them back*, and *what becomes possible*.

This review is organized around those four questions—collaboration, barriers, archetypes, and opportunities—and is written to evaluate a specific hypothesis supplied by the reader: that the future of science “feels like a big group chat of the smartest minds glued together by frontier models.” That image makes two distinct claims worth separating. The first is a claim about *connectivity*: that frontier models act as the glue binding people and ideas together. The second is a claim about *sociality*: that the resulting mode of work is a group chat, i.e., a densely social, many-to-many conversation. We return to both in the discussion (Section 8), and we find the literature treats them very differently.

1.2 Scope and method

We surveyed empirical surveys of researchers, large-scale bibliometric studies, HCI and CSCW qualitative work, science-of-science analyses, autonomous-agent system papers, and the policy and position literature, spanning the period from early 2023 through mid-2026. Sources were gathered through structured multi-angle literature search and each citation’s metadata (authors, venue, year, DOI or arXiv identifier) was verified against the primary record where possible; grey literature (publisher survey reports) and non-peer-reviewed preprints are flagged as such in context and in Section 9. The review is interpretive rather than a PRISMA-style systematic review: our aim is a faithful map of what has been claimed and how well it is evidenced, not an exhaustive enumeration. Section 9 states the coverage limits candidly.

1.3 A note on terminology

We use “LLM” and “generative AI” interchangeably for the general-purpose conversational and text-generation models (GPT-4 and successors, Gemini, Claude, and open-weight equivalents) that dominate the studied period. Where a source concerns narrow or task-specific machine learning rather than generative models we say so, because several bibliometric studies of “AI in science” (Gao and Wang, 2024; Hao et al., 2025) measure a broader category than chat-based LLMs alone.

2 Adoption: Who Is Using LLMs, and for What

Any account of LLMs in research has to begin with the sheer speed and breadth of uptake, because the scale of adoption reframes every subsequent question. Three complementary evidence streams—reader and community surveys, publisher survey programs, and population-scale text forensics—converge on the same conclusion: LLM use became the majority condition among active researchers within roughly two years, but it is concentrated in a narrow band of low-risk tasks.

2.1 Survey evidence

The earliest signal came from Nature’s reader questionnaire in early 2023, in which about 80% of 672 respondents had used ChatGPT or a similar tool and more than 20% used one regularly (Owens, 2023). Nature’s global postdoc survey later that year found roughly one in three employed postdocs using AI chatbots to refine text, generate or edit code, and manage literature, with 43% of chatbot users doing so weekly (Nordling, 2023). By the 2025 PhD survey, 53% of candidates reported weekly AI use (Nordling, 2025).

Publisher survey programs tell a steeper story. Wiley’s multi-wave *ExplanaItions* study recorded overall AI-tool usage rising from 57% of researchers in 2024 to 84% in 2025, and use for research and publication tasks specifically from 45% to 62% in the same interval, with 85% reporting efficiency gains (Wiley, 2025; Naddaf, 2025a). Oxford University Press found 76% of 2,345 researchers already using some AI tool, most commonly machine translation and chatbots (Oxford University Press, 2024). Elsevier’s survey of about 3,000 researchers and clinicians across 123 countries found high expectations but, at the time of fielding, still-modest actual work adoption (Elsevier, 2024). These are self-selected, non-probability samples run by interested parties, and should be read as directional grey literature rather than population estimates—but their trajectories agree.

2.2 Bibliometric forensics

The most rigorous adoption evidence does not ask researchers what they do; it measures the residue in the literature itself. Liang et al. (2025) analyzed roughly 1.12 million preprints and papers across arXiv, bioRxiv, and the Nature portfolio (2020–2024) and estimated that LLM-modified sentences reached up to 22% in computer-science abstracts, versus up to about 9% in mathematics and Nature-portfolio papers. Working from a different method—a surge in “excess” stylistic vocabulary—Kobak et al. (2025) examined more than 15 million PubMed abstracts and concluded that at least 13.5% of 2024 biomedical abstracts were LLM-processed, exceeding 40% in some subcorpora. A more conservative keyword-fingerprint estimate over the Dimensions database put LLM-assisted output at slightly above 1% of 2023 articles, or roughly 60,000 papers (Gray, 2024); the gap between estimates reflects genuine methodological uncertainty about detection thresholds, discussed in Section 9.

2.3 The shape of usage

Two features of the usage distribution recur across sources and matter for the rest of the review. First, usage is *task-stratified*: it is highest for editing, translation, coding, and literature triage, and lowest—often actively disapproved—for authorship of scientific claims and for peer review (Kwon, 2025; Elsevier, 2024). Second, usage is *uneven across people*, with gradients by field, career stage, gender, and workplace that Section 6 treats in detail. The headline is not that everyone uses LLMs the same way; it is that most researchers now use them for some things and few use them for everything.

3 The Interaction Surface: Subscriptions, Assistants, and Agents

A dimension that the adoption figures in Section 2 flatten is *which surface* the researcher actually touches. “Using an LLM” spans at least three materially different modes: a consumer chat subscription (ChatGPT, Claude, Gemini) accessed by copy-and-paste; an IDE-integrated coding assistant (GitHub Copilot, Cursor) offering inline completion; and an autonomous agent that plans and executes multi-step work with limited supervision. These surfaces differ in cost, in trust dynamics, in who adopts them, and—most relevant to the group-chat hypothesis—in whether the researcher is talking *to* a model or *delegating to* a team of them. This section treats the distinction directly.

3.1 Consumer chat leads; IDE-integrated tools trail

Where usage can be broken out by surface, general-purpose chat dominates. The 2025 Stack Overflow Developer Survey found ChatGPT used by 81.7% of AI-using developers versus 67.9%

for GitHub Copilot, 47.4% for Gemini, and 40.8% for Claude Code, against an overall AI-adoption rate of 84% (Stack Overflow, 2025). JetBrains’ independent survey of 23,262 developers showed the same ordering with a roughly 2x gap: ChatGPT tried by 69% and used regularly by 49%, versus Copilot tried by 40% and used regularly by 26% (JetBrains, 2024). The 2024 Stack Overflow wave established the baseline, with 62% actively using AI tools and nearly identical adoption between professional developers (63.2%) and those learning to code (63.0%) (Stack Overflow, 2024).

Among researchers specifically the chat-forward pattern is, if anything, sharper. O’Brien et al.’s survey of 868 scientists who program found an overwhelming preference for general-purpose conversational interfaces over developer-specific, IDE-integrated tools—the opposite of mainstream software-engineering recommendation—with adoption highest among students and less-experienced programmers, and with the number of generated lines accepted at once the strongest predictor of *perceived* productivity (O’Brien et al., 2025). A nationally representative survey of 232 PhD-level scientists at US R1 universities found 65% had used generative AI in teaching or research, using it in research for writing and editing (40%), data analysis (32%), and conceptualization (28%) (Arroyo-Machado et al., 2025). A CHI 2025 mixed-methods study inside a national laboratory captured the surface split qualitatively, organizing scientists’ use into “AI as copilot” (interactive coding and writing help) and “AI as workflow agent” (automating multi-step tasks)—the two poles this section connects (Wagman et al., 2025).

3.2 The productivity contradiction

The evidence on whether these tools actually make technical work faster is genuinely contradictory, and the contradiction tracks the surface and the user. The canonical positive result is a controlled experiment in which developers using GitHub Copilot completed an HTTP-server task 55.8% faster than a control group, with the largest gains among less-experienced developers (Peng et al., 2023). The canonical negative result is METR’s 2025 randomized trial, in which 16 experienced open-source developers working on mature repositories were *slowed by 19%* when given AI tools (primarily Cursor Pro with Claude), even though they predicted a 24% speedup and still believed they had been sped up afterward (Becker et al., 2025). Field studies help reconcile the two: a workplace diary study found sustained use raised developers’ sense of usefulness and enjoyment while trust in output correctness did *not* move (Butler et al., 2024), and a longitudinal case study of 26,317 commits found Copilot adopters were already more active before adoption (self-selection) and showed no statistically significant change afterward despite perceiving gains (Stray et al., 2026). The through-line—perceived productivity decoupled from measured productivity—is exactly the calibration illusion that Section 5 flagged, and it is most dangerous for the inexperienced, chat-forward users who predominate among scientist-programmers (O’Brien et al., 2025).

3.3 The autonomy spectrum: assistant to agent

The sharpest surface distinction is between an assistant a researcher prompts and an agent that acts. Benchmarks chart the movement along this spectrum. SWE-bench posed 2,294 real GitHub issues and found the best model of early 2024, Claude 2, resolved just 1.96%—a stark baseline for autonomous coding (Jimenez et al., 2024). Purpose-built agent scaffolding changed the picture fast: SWE-agent’s agent-computer interface lifted GPT-4 Turbo to 12.5% on the same benchmark, showing that *interface design*, not only model quality, governs agentic capability (Yang et al., 2024). From there the autonomous end of the spectrum is the same systems that anchor Section 4: The AI Scientist-v2 running a self-improving experiment loop to a workshop-accepted paper (Yamada et al., 2025), AlphaEvolve evolving code to beat a 56-year-old algorithmic bound (Novikov et al.,

2025), Agent Laboratory reporting an 84% cost reduction while finding that human feedback at each stage improves quality (Schmidgall et al., 2025), and Robin driving a discovery loop to a validated drug-repurposing candidate (Ghareeb et al., 2025).

The crucial qualification is that autonomy degrades with task difficulty in a measurable way. A study of *mergen*, an R package delegating genomics data analysis to GPT-3.5 and GPT-4, found correct-code rates collapsing from 88% at low complexity to 0% at the highest, even with self-correction (Jansen et al., 2025). This is the empirical shape of the “jagged frontier” (Section 6) rendered in code: autonomous agents work where the task is well-instrumented and bounded, and fail as it opens up. The assistant surface keeps a human in every loop; the agent surface removes them from most—and the evidence says removing them is safe only inside the frontier’s boundary.

3.4 Access economics and the freemium divide

Finally, the surfaces differ in who can afford them, which converts a product distinction into an equity one. Despite near-universal adoption, willingness to pay is low: in a survey of an elite US college, only 11.4% of generative-AI users paid for any service and 88.0% relied on free tiers, with payment skewed by gender and demographics—leading the authors to warn of a new “freemium” inequality if paid tiers (faster, more capable frontier models) confer academic advantage that lower-income users cannot buy (Contractor and Reyes, 2025). This compounds the reproducibility objection from Section 5: paid, closed, frontier models are exactly the ones that are versionless and deprecatable, so the most capable surface is also the least reproducible (Palmer et al., 2024). Institutional provisioning mediates the gap unevenly—JetBrains found nearly 80% of organizations either permit third-party AI tools or have no policy at all (JetBrains, 2024)—meaning access is increasingly set by whether a researcher’s institution pays, not by what the research needs.

3.5 Implication for the group-chat hypothesis

The surface distinction refines the discussion in Section 8. A consumer chat subscription is structurally a *private, one-to-one channel*: the mode most scientists actually use, and the one furthest from a group chat. The genuinely multi-party, “room full of minds” surface is the autonomous multi-agent system—but that surface is still the province of specialist builders and narrow, well-instrumented problems, not the everyday researcher. The gap between the surface researchers use (private chat) and the surface that would realize the hypothesis (agent teams) is itself part of why the group-chat future remains aspirational rather than observed.

4 LLMs in Research Collaboration

Collaboration is the theme most directly implicated by the reader’s group-chat hypothesis, and the literature approaches it at three levels: the LLM as a partner to an individual researcher, the LLM as one or many autonomous agents in a research team, and the LLM’s aggregate effect on the human collaboration networks of science.

4.1 Human–AI teaming in scientific work

A body of HCI and organizational research reframes the LLM from tool to teammate. Lou et al. (2025) synthesize this shift through the lens of Team Situation Awareness theory, arguing that genuine partnership requires shared mental models and calibrated trust rather than mere access, and proposing a four-dimension framework (formulation, coordination, maintenance, training) for

human–AI teams. Autonomy-level frameworks give this a vocabulary: Morris et al. (2024) define six “Levels of Autonomy” from AI-as-tool to AI-as-agent, explicitly analogized to self-driving levels, and Gao et al. (2024) adapt the same idea to biomedicine, positing a ladder from machine-learning-as-tool (Level 0) through AI as hypothesis-refining collaborator (Level 2) to AI as scientist (Level 3). These frameworks matter because they let the field talk precisely about *how much* of the scientific act is delegated, which is exactly the variable that the barriers and archetypes literatures show researchers calibrating.

Qualitative studies show what teaming looks like in practice, and it is messier than the frameworks suggest. Interview studies of scientists using LLMs for literature reviews surface a persistent “verification burden”: outputs cannot be trusted without checking, and the checking often costs as much as the task saved (Nogueira et al., 2025). Studies of scientific writing assistance find that perception of AI involvement can matter more than actual quality—in a randomized trial, reviewers’ acceptance tracked how much an abstract had been edited rather than whether a human or an LLM wrote it (Hazra et al., 2026). Purpose-built collaborative systems such as SciSciGPT, an open-source multi-agent assistant for the science of science, illustrate the constructive end of this work: the LLM retrieves data, runs analyses, and proposes hypotheses as a partner in an empirical workflow (Shao et al., 2025).

4.2 Multi-agent and autonomous “AI scientist” systems

The most visible collaboration research imagines the LLM not as a single partner but as an entire research team. Early demonstrations showed LLM agents closing the loop from reasoning to physical experiment: Coscientist coupled GPT-4 with web search, documentation, code execution, and robotic hardware to plan and execute real cross-coupling reactions (Boiko et al., 2023), and ChemCrow augmented GPT-4 with eighteen expert tools to plan and, on a robotic platform, execute syntheses (M. Bran et al., 2024).

The frontier then moved to end-to-end automation of the research act itself. Sakana AI’s *The AI Scientist* generates ideas, writes and runs experiment code, produces plots, drafts a full paper, and runs a simulated review, at under \$15 per paper (Lu et al., 2024). Its successor removed the dependence on human-authored code templates via an agentic tree search and reported that, of three fully AI-generated manuscripts submitted to an ICLR workshop, one scored above the average human acceptance threshold—described as the first AI-generated paper to pass peer review (Yamada et al., 2025). Notably, the framing of Lu et al. (2024) is explicitly social: the system is meant to iterate open-endedly, “acting like the human scientific community”—a direct textual echo of the group-chat image this review evaluates.

Multi-agent architectures push furthest toward the collaboration metaphor. Google’s AI co-scientist uses a generate–debate–evolve loop over many Gemini-based agents with an Elo-style tournament, and recapitulated an unpublished mechanism of antimicrobial-resistance gene transfer (Gottweis et al., 2025). The Virtual Lab casts an LLM “Principal Investigator” agent leading specialist LLM agents through structured meetings with light human oversight; the human contributed only about 1% of the conversation, yet the team designed 92 nanobodies, of which validated binders to recent SARS-CoV-2 variants emerged (Swanson et al., 2025). FutureHouse’s agents extend the pattern to literature and discovery: PaperQA2 matches or exceeds experts at retrieval, summarization, and contradiction detection (Skarlinski et al., 2024); Aviary trains small agents that rival frontier models at up to 100x lower cost (Narayanan et al., 2024); and Robin drove the hypothesis–experiment–analysis loop to nominate a novel candidate treatment for dry age-related macular degeneration (Ghareeb et al., 2025). Agent Laboratory frames the same machinery as an assistant rather than a replacement, reporting that intermediate human feedback significantly improves out-

put quality (Schmidgall et al., 2025), and AlphaEvolve, an evolutionary coding agent orchestrating Gemini models, improved on Strassen’s 56-year-old matrix-multiplication result (Novikov et al., 2025). A recent survey frames this trajectory as a shift from “AI for Science” to “Agentic Science” (Wei et al., 2025).

These systems are the strongest evidence for the “glued together by frontier models” half of the hypothesis: they show frontier models acting as connective tissue not just between a researcher and their tools but between many role-specialized agents. Section 5 supplies the essential counterweight—critical evaluations showing the outputs are frequently shallow, error-prone, or over-claimed—and Section 8 weighs the two.

4.3 LLMs and the structure of scientific teams

The third level asks what LLMs do to the human collaboration network in aggregate, and here the most important finding in this review appears. Hao et al. (2025), analyzing 41.3 million natural-science papers, find that AI-augmented researchers publish 3.02x more papers, earn 4.84x more citations, and reach project-leadership roles 1.37 years earlier—but that AI adoption *shrinks* the collective volume of topics studied by 4.63% and reduces scientist-to-scientist engagement by 22%, a pattern the authors label “lonely crowds.” Gao and Wang (2024) similarly link AI use across roughly 75 million publications to higher hit-paper rates while noting a large gap between AI education and application. The juxtaposition is stark: at the individual level LLMs look like an amplifier of reach and impact, but at the network level the current evidence points toward *fewer* human ties, not the denser many-to-many sociality the group-chat metaphor implies. We take this tension up directly in Section 8.

5 Barriers to Use

Adoption has outrun the norms, tooling, and evidence needed to make it safe, and the literature on barriers is correspondingly the largest and most contested part of this corpus. We group it into four clusters: accuracy and trust; policy and integrity; equity and access; and epistemic risk. Several barriers are not merely frictions to be engineered away but structural features of how closed generative models interact with the norms of science.

5.1 Accuracy, hallucination, and trust

The founding technical barrier is fabrication. An early case report coined “artificial hallucination” for ChatGPT’s production of plausible but invented medical claims and non-existent references (Alkaissi and McFarlane, 2023). The definitive quantitative study analyzed 636 citations in ChatGPT-generated literature reviews and found 55% of GPT-3.5 citations and 18% of GPT-4 citations wholly fabricated, with substantive errors in many of the rest—and fabricated references often carry real journal names, making them deceptively plausible (Walters and Wilder, 2023). This is why trust, not access, is the operative variable: Nature’s 2025 PhD survey found 81% of candidates do not fully trust AI tools even as 53% use them weekly (Nordling, 2025), and a survey at an R1 university found trust and adoption significantly correlated with fact-checking as the dominant concern (Jung et al., 2025). Qualitative work shows this trust is not binary but continuously calibrated against expectations, expertise, and stakes (Ryser et al., 2025), and that in practice it manifests as the verification burden documented for literature reviews (Nogueira et al., 2025).

5.2 Policy, authorship, and research integrity

The policy landscape is fragmented and, in places, prohibitive. Publishers converged quickly on one point—no LLM can be an author, because authorship implies accountability an LLM cannot bear—stated by Nature (Nature Editorial, 2023), COPE (Committee on Publication Ethics, 2023), and ICMJE (International Committee of Medical Journal Editors, 2023), while Science initially banned AI-generated text outright before relaxing to disclosure (Thorp, 2023). Funders went further on the evaluation side: both NIH (National Institutes of Health, 2023) and NSF (U.S. National Science Foundation, 2023) prohibit reviewers from uploading proposal content to generative AI tools on confidentiality grounds. Yet enforcement lags badly. Liang et al. (2024) estimate that 6.5–16.9% of text in reviews at major AI conferences after ChatGPT’s release was substantially LLM-modified, concentrated in low-confidence, deadline-rushed reviews; a Frontiers survey found more than half of academics have used AI in peer review, often against guidance (Naddaf, 2025b); and quasi-experimental work suggests AI-assisted reviews raise paper scores and borderline acceptance rates, an “AI review lottery” (Russo et al., 2025). A randomized study delivering LLM feedback across more than 20,000 ICLR 2025 reviews found 27% of reviewers updated their reviews in response (Thakkar et al., 2026). Meanwhile the integrity literature documents the downside at scale: a record 10,000-plus retractions in 2023 driven by paper mills whose economics generative AI improves (Van Noorden, 2023), and 139 GPT-fabricated papers already indexed in Google Scholar, most on policy-relevant topics (Haider et al., 2024). Nature’s 2025 poll of about 5,000 researchers captured the resulting normative disarray: over 90% accept AI editing, first-drafting is contested, and AI-written peer review is widely condemned (Kwon, 2025).

A quieter but structurally important barrier is reproducibility. Palmer et al. (2024) argue that closed, versionless, deprecatable commercial models undermine scientific provenance—a result obtained with a model that no longer exists cannot be reliably re-run—and that proprietary-model use should require explicit justification. Governance proposals such as “living guidelines” with independent oversight occupy the middle ground between bans and laissez-faire disclosure (Bockting et al., 2023).

5.3 Equity, access, and skills

Whether LLMs widen or narrow inequities in science is genuinely two-sided. The optimistic case starts from a documented baseline harm: non-native English speakers pay large time and rejection penalties in a English-dominated publishing system (Amano et al., 2023), and low-cost AI proofreading and translation could level that field (Berdejo-Espinola and Amano, 2023). The pessimistic case is that enforcement tooling reintroduces the bias: GPT detectors misclassify writing by non-native English speakers as AI-generated at very high rates, creating a double bind (Liang et al., 2023). Adoption itself is stratified: Dorta-González et al. (2024) find usage roughly 7% lower among women, about 19% lower among senior researchers, and depressed by lack of training, while for-profit and medical settings raise it. Environmental and cost objections form a further access barrier, with inference-phase energy for general-purpose generative models measured at orders of magnitude above task-specific alternatives (Luccioni et al., 2024).

5.4 Epistemic risks

The deepest barrier is not that LLMs err but that they may systematically distort scientific knowing even when they perform well. Messeri and Crockett (2024) argue that AI tools produce “illusions of understanding”—researchers believe they grasp more than they do—and risk scientific monocultures that “produce more but understand less.” Empirical work gives the homogenization worry teeth:

generative AI raised individual creativity but reduced the collective diversity of outputs in a large writing experiment (Doshi and Hauser, 2024), and LLM-assisted ideation converged different people onto semantically similar ideas (Anderson et al., 2024). At the level of the literature itself, the excess-vocabulary signal documents stylistic convergence across millions of abstracts (Kobak et al., 2025), and federal-funding analysis links higher LLM involvement in proposals to reduced semantic distinctiveness (Qian et al., 2026). Finally, a controlled EEG study reported that essay-writing with an LLM produced weaker neural connectivity, reduced ownership, and impaired recall—“cognitive debt”—persisting after the tool was withdrawn (Kosmyna et al., 2025); as a non-peer-reviewed preprint with a small sample it should be read cautiously, but it crystallizes the deskilling concern that runs through the survey data on eroded critical thinking (Nordling, 2025; Oxford University Press, 2024).

6 Researcher Archetypes

The adoption data in Section 2 describe an average; the archetypes literature describes the distribution around it. Two kinds of segmentation appear: demographic (who adopts, by observable attributes) and behavioral (how they work with the model once they do). The behavioral typologies are the more theoretically interesting, because they predict *outcomes*, not just uptake.

6.1 Demographic and disciplinary segmentation

Field is the strongest axis. Van Noorden and Perkel’s survey of 1,600-plus researchers found usage heaviest in computer science, engineering, and mathematics and lightest in the humanities, with the bibliometric fingerprint studies confirming the gradient—up to 22% LLM-modified abstracts in CS versus around 9% in mathematics (Van Noorden and Perkel, 2023; Liang et al., 2025). Career stage is the second axis, and its direction is consistent: early-career researchers adopt more readily and senior researchers less, with Dorta-González et al. (2024) estimating roughly 19% lower usage among senior researchers and OUP finding early-career researchers both heavier users and, in some respects, more skeptical (Oxford University Press, 2024). Gender, training access, and workplace type (for-profit and medical settings raising usage) round out the demographic picture (Dorta-González et al., 2024). Among scientists who program, adoption is highest among students and less-experienced programmers, who favor general-purpose chat interfaces over developer tools—and, strikingly, those with less software-engineering discipline report *higher* perceived productivity, a possible calibration illusion (O’Brien et al., 2025).

6.2 Behavioral typologies: centaurs, cyborgs, and self-automators

The most useful archetypes come from field-experimental work on knowledge workers, transferable to researchers. Dell’Acqua et al. (2026), in a preregistered experiment with 758 BCG consultants, established the “jagged technological frontier”: AI sharply improves performance on tasks inside its capability boundary (12.2% more tasks, 40% higher quality) but degrades it outside. They distinguished *centaurs*, who divide labor strategically between human and AI at the sub-task level, from *cyborgs*, who interleave the two continuously. A follow-up field study refined this into three empirical modes: *cyborgs* (around 60%, fused co-creation across the workflow), *centaurs* (directed co-creation, treating AI as a search engine or writing assistant), and *self-automators* (abdicated co-creation) (Randazzo et al., 2025). Crucially, the modes predict skill trajectories: centaurs achieved the highest accuracy and *strengthened* their domain expertise, whereas self-automators grew neither domain nor AI expertise. This maps directly onto the deskilling and verification-burden findings

in Section 5: the archetype that keeps a human firmly in the loop is the one that both performs better and avoids cognitive debt.

6.3 Attitudinal archetypes and the trust–use gap

A third segmentation is attitudinal. Across surveys a recurring split separates enthusiastic adopters, pragmatic-but-wary majority users, and principled abstainers, cross-cut by a persistent trust–use gap: many researchers use tools they say they do not trust (Nordling, 2025; Jung et al., 2025). Qualitative work in specific research communities sharpens this. In HCI, scholars articulate ethical concerns about LLMs but rarely implement safeguards—an awareness–action gap (Kapania et al., 2024). Among qualitative researchers, LLMs provoke boundary-defense: using models as synthetic interview participants is judged to foreclose consent and delegitimize the method (Kapania et al., 2025; Schroeder et al., 2025). An “insider–outsider” analysis argues that LLMs simultaneously empower newcomers and threaten established expertise, reshaping who counts as a knowledge producer (Xu et al., 2025). The through-line is that archetype is not fixed by demography; it is a stance toward delegation that each researcher negotiates against their field’s norms.

7 Opportunities for Innovation

Where the barriers literature is cautionary, a parallel body of work maps what LLMs make newly possible. We organize it by the stage of the research lifecycle each opportunity touches, and throughout we pair the optimistic claim with its best empirical check, because the most interesting finding in this area is how often a capability that looks strong at one stage evaporates at the next.

7.1 Ideation and hypothesis generation

The provocative result is that LLMs can generate research ideas judged *more novel* than expert humans’. In a controlled study recruiting more than 100 NLP researchers, blind reviewers rated LLM-generated ideas significantly more novel than expert ideas, though slightly less feasible (Si et al., 2024). The essential corrective is the same authors’ execution study: when 43 experts each spent over 100 hours turning a randomly assigned idea into a real paper, LLM-idea scores dropped significantly more than human-idea scores on every metric, erasing and in places reversing the novelty advantage (Si et al., 2025). The lesson is not that LLM ideation is worthless but that ideation-stage novelty is a poor predictor of executed value—an “ideation–execution gap” that any group-chat vision of frontier-model brainstorming must reckon with.

7.2 Literature synthesis and search

Literature work is where agentic systems look most immediately useful. PaperQA2 matches or exceeds subject-matter experts at retrieval, summarization, and contradiction detection, and writes cited summaries more accurate than existing human-written reference articles (Skarlinski et al., 2024). Yet user studies of scientists doing real literature reviews find the bottleneck is not generation but trust: the verification burden and tool fragmentation dominate the experience, motivating design goals around step-by-step verification and explainable, source-grounded output (Nogueira et al., 2025). The opportunity is real but conditioned on making the output auditable.

7.3 Writing, coding, and analysis

This is the stage with the broadest, most-accepted use. Surveys put editing and translation at over 90% acceptability (Kwon, 2025), and empirical assessments show LLMs handling application code and data-analysis scripts, albeit with tool-dependent reliability and recurring integrity issues (Nejjar et al., 2025). For non-native English speakers specifically, writing assistance is the clearest equity opportunity in the entire corpus (Berdejo-Espinola and Amano, 2023; Amano et al., 2023). The caution from Section 5 applies in proportion to stakes: low for language editing, higher for analysis whose correctness the researcher cannot independently verify.

7.4 Peer review and evaluation

LLMs are already reshaping evaluation whether or not policy permits it. Beyond the prevalence findings, a randomized study showed LLM-generated feedback caused 27% of reviewers to update their reviews, incorporating over 12,000 suggestions (Thakkar et al., 2026), suggesting a constructive assistive role distinct from wholesale delegation. The opportunity here is genuinely double-edged: the same capability that can raise review quality also enables the confidentiality breaches and score inflation that funders and studies warn against (National Institutes of Health, 2023; Russo et al., 2025).

7.5 Autonomous discovery

The frontier opportunity is closing the loop entirely, and the evidence is now concrete rather than speculative. Multi-agent systems have produced experimentally validated results—nanobody binders (Swanson et al., 2025), a recapitulated antimicrobial-resistance mechanism (Gottweis et al., 2025), a nominated macular-degeneration therapy (Ghareeb et al., 2025)—and AlphaEvolve improved a 56-year-old algorithmic bound (Novikov et al., 2025). But the critical evaluations are equally concrete: a detailed assessment of The AI Scientist found 42% of its experiments failed on coding errors, a median of five mostly outdated citations per paper, and outputs resembling “a rushed undergraduate paper” (Beel et al., 2025). The honest reading is that autonomous discovery works today in narrow, well-instrumented domains with tight human oversight and validation, and fails or over-claims when pushed toward open-ended, unsupervised science—a boundary that maps neatly onto the jagged frontier of Section 6.

8 Discussion: Is the Future of Science a Group Chat?

We can now evaluate the reader’s hypothesis—that the future of science “feels like a big group chat of the smartest minds glued together by frontier models” —against the assembled evidence. As noted in Section 1, the metaphor bundles two separable claims, and the literature treats them very differently.

8.1 The “glue” claim is well supported

That frontier models are becoming the connective tissue of research is the best-evidenced proposition in this review. At the individual level, LLMs already sit between the researcher and nearly every artifact they touch: the literature (Skarlinski et al., 2024), code and data (Nejjar et al., 2025), drafts (Kwon, 2025; Hazra et al., 2026), and reviews (Thakkar et al., 2026). At the system level, multi-agent architectures make the gluing literal, binding role-specialized agents—principal investigator, critic, analyst—into teams that have generated experimentally validated discoveries (Swanson et al.,

2025; Gottweis et al., 2025; Ghareeb et al., 2025). The framing of Lu et al. (2024), in which the system iterates “like the human scientific community,” is the hypothesis stated almost verbatim by the builders themselves. If the claim is that frontier models are becoming the substrate through which scientific work is coordinated, the evidence is firmly on the reader’s side.

8.2 The “group chat” claim faces a direct empirical challenge

The second claim—that the resulting mode is a densely social, many-to-many conversation—runs into the strongest counter-evidence in the corpus. The group-chat image predicts *more* connection: more minds, more cross-talk, a wider conversation. The best science-of-science measurement to date finds the opposite at the human layer. Hao et al. (2025), across 41.3 million papers, report that AI-augmented researchers reduce scientist-to-scientist engagement by 22% and that AI adoption contracts the collective range of topics by 4.63%—“lonely crowds,” not a group chat. The homogenization results compound this: LLMs raise individual output while converging ideas toward each other (Doshi and Hauser, 2024; Anderson et al., 2024; Qian et al., 2026), and Messeri and Crockett (2024) warn this produces monocultures that understand less even as they produce more. The picture is less a room full of the smartest minds talking than a set of powerful *private* channels, each pairing one researcher with a model.

8.3 Reconciling the two

The tension resolves if we are precise about *who* is in the chat. The sociality the literature documents is between *humans and models*, not among *humans*. Frontier models are indeed gluing the smartest *ideas* together—distilled from the training corpus and the live literature—but the current empirical signature is substitution of human collaboration by model collaboration, not augmentation of it. Whether that is a transient artifact of an immature tooling era or a durable property is the open question. The archetypes evidence offers a hopeful hedge: the “centaur” mode, which keeps humans structurally in the loop, both outperforms and preserves expertise (Dell’Acqua et al., 2026; Randazzo et al., 2025), suggesting the lonely-crowd outcome is a *choice* about how the tools are used rather than an inevitability of the technology. A group chat of the smartest minds is a plausible *design target*; it is not yet the observed equilibrium.

8.4 What would confirm the hypothesis

If the reader’s vision is right, we would expect future science-of-science studies to reverse the Hao et al. (2025) finding—rising human co-authorship and topic diversity *alongside* heavy model use—and to show multi-agent systems that broaden rather than narrow the questions asked. The current trajectory does not yet show this, but the field is barely three years old and the measurement instruments are new. The honest synthesis is that the reader has identified the right *mechanism* (frontier models as glue) and an aspirational *social form* (the group chat) that the evidence has not yet delivered—and may not, absent deliberate design for it.

9 Limitations of This Review

Several limitations bound the confidence of the conclusions above.

Coverage. This is an interpretive review, not a systematic one. It draws on roughly seventy sources selected through structured multi-angle search rather than an exhaustive database protocol,

so relevant work—particularly in non-English venues, in disciplines outside STEM and HCI, and in the fast-moving 2026 preprint stream—is certainly missing. The claim implicit in the reader’s brief, that this covers “95% of the literature,” should be read as an aspiration toward representativeness of the major *findings and debates*, not as a census; the field now produces relevant papers faster than any review can close over them.

Evidence quality is uneven. The corpus mixes peer-reviewed studies, industry survey reports, and non-peer-reviewed preprints. Publisher surveys (Wiley, OUP, Elsevier) use self-selected samples and are produced by parties with a commercial interest in the answer; they are treated as directional grey literature. Several load-bearing preprints—including the “cognitive debt” EEG study (Kosmyna et al., 2025), the federal-funding analysis (Qian et al., 2026), and several HCI and teaming papers—had not completed peer review at the time of writing and are flagged as such in context.

Measurement uncertainty in the bibliometrics. The population-scale usage estimates rest on stylometric detection whose thresholds are contested: the same underlying reality yields “at least 13.5%” (Kobak et al., 2025), “up to 22%” (Liang et al., 2025), and “slightly over 1%” (Gray, 2024) depending on method and corpus. GPT detectors are known to be biased (Liang et al., 2023). These numbers indicate direction and rough magnitude, not precision.

Pace of change. Every capability claim is a snapshot of a moving target. The autonomous-discovery results in particular reflect specific model generations and will date quickly; the critical evaluations (Beel et al., 2025) are as time-bound as the systems they critique. The discussion in Section 8 deliberately frames the group-chat question as an *open trajectory* rather than a settled outcome for this reason.

Metadata caveats. Citation metadata was verified against primary records where accessible, but a minority of entries retain unresolved details (truncated author lists on large-collaboration papers, forthcoming volume and page numbers for 2025–2026 articles, and venue confirmation for recently accepted conference papers). These are noted in the bibliography source and do not affect the substantive claims.

10 Conclusion

Across roughly seventy sources, a consistent picture emerges. LLM use in research is now the majority condition, but it is concentrated in low-risk, high-leverage tasks—editing, translation, coding, literature triage—and thins out sharply as the stakes and the required trust rise, reaching its lowest social acceptance at peer review and authorship. The barriers are real and structural, not merely teething problems: hallucinated citations, an incoherent policy patchwork, reproducibility concerns with closed models, measurable demographic inequities in access, and a credible if unsettled case that AI assistance homogenizes ideas and manufactures illusions of understanding. The archetypes literature shows this is not a uniform population but a stratified one, with the most durable gains accruing to researchers who keep a human hand firmly on the process—the “centaurs” rather than the “self-automators.”

On the reader’s hypothesis, the verdict is split and, we think, genuinely informative. Frontier models are indeed becoming the glue of research: they connect a scientist to the literature, to code, to collaborators’ drafts, and—in their multi-agent form—to teams of specialist agents that have

already produced experimentally validated results (Swanson et al., 2025; Gottweis et al., 2025). But the “group chat” half of the metaphor runs into the strongest single piece of counter-evidence in this review: at population scale, AI-augmented scientists so far collaborate with *fewer* humans, not more, and the collective conversation of science *narrows* rather than widens (Hao et al., 2025). The future the literature actually sketches is less a group chat than a set of powerful private channels—each researcher paired with a model that amplifies their reach while, for now, quietly thinning the room.

References

- Hussam Alkaissi and Samy I. McFarlane. Artificial hallucinations in ChatGPT: Implications in scientific writing. *Cureus*, 15(2):e35179, 2023. doi: 10.7759/cureus.35179. URL <https://doi.org/10.7759/cureus.35179>.
- Tatsuya Amano, Valeria Ramírez-Castañeda, Violeta Berdejo-Espinola, Israel Borokini, Shawan Chowdhury, Marina Golivets, Juan David González-Trujillo, Flavia Montaña-Centellas, Kumar Paudel, Rachel Louise White, and Diogo Veríssimo. The manifold costs of being a non-native English speaker in science. *PLOS Biology*, 21(7):e3002184, 2023. doi: 10.1371/journal.pbio.3002184. URL <https://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.3002184>.
- Barrett R. Anderson, Jash Hemant Shah, and Max Kreminski. Homogenization effects of large language models on human creative ideation. In *Proceedings of the 16th Conference on Creativity and Cognition (C&C)*, pages 413–425, 2024. doi: 10.1145/3635636.3656204. URL <https://doi.org/10.1145/3635636.3656204>.
- Wenceslao Arroyo-Machado, Jinghuan Ma, Tipeng Chen, Timothy P. Johnson, Shaika Islam, Lesley Michalegko, and Eric W. Welch. Generative AI and academic scientists in US universities: Perception, experience, and adoption intentions. *PLOS ONE*, 20(8):e0330416, 2025. doi: 10.1371/journal.pone.0330416. URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0330416>.
- Joel Becker, Nate Rush, Elizabeth Barnes, and David Rein. Measuring the impact of early-2025 AI on experienced open-source developer productivity, 2025. URL <https://arxiv.org/abs/2507.09089>. METR randomized controlled trial.
- Joeran Beel, Min-Yen Kan, and Moritz Baumgart. Evaluating Sakana’s AI scientist: Bold claims, mixed results, and a promising future? *ACM SIGIR Forum*, 2025. doi: 10.1145/3769733.3769747. URL <https://arxiv.org/abs/2502.14297>.
- Violeta Berdejo-Espinola and Tatsuya Amano. AI tools can improve equity in science. *Science*, 379(6636):991, 2023. doi: 10.1126/science.adg9714. URL <https://www.science.org/doi/10.1126/science.adg9714>.
- Claudi L. Bockting, Eva A. M. van Dis, Robert van Rooij, Willem Zuidema, and Johan Bollen. Living guidelines for generative AI — why scientists must oversee its use. *Nature*, 622:693–696, 2023. doi: 10.1038/d41586-023-03266-1. URL <https://www.nature.com/articles/d41586-023-03266-1>.

- Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023. doi: 10.1038/s41586-023-06792-0. URL <https://www.nature.com/articles/s41586-023-06792-0>.
- Jenna Butler, Jina Suh, Sankeerti Haniyur, and Constance Hadley. Dear diary: A randomized controlled trial of generative AI coding tools in the workplace. 2024. doi: 10.48550/arXiv.2410.18334. URL <https://arxiv.org/abs/2410.18334>.
- Committee on Publication Ethics. Authorship and AI tools: COPE position statement. COPE Council, February 2023. URL <https://publicationethics.org/guidance/cope-position/authorship-and-ai-tools>. Published 13 February 2023.
- Zara Contractor and Germán Reyes. Generative AI in higher education: Evidence from an elite college. Working paper, arXiv, 2025. URL <https://arxiv.org/abs/2508.00717>.
- Fabrizio Dell’Acqua, Edward McFowland III, Ethan R. Mollick, Hila Lifshitz-Assaf, Katherine C. Kellogg, Saran Rajendran, Lisa Kraye, François Candelon, and Karim R. Lakhani. Navigating the jagged technological frontier: Field experimental evidence of the effects of artificial intelligence on knowledge worker productivity and quality. *Organization Science*, 37(2):403–423, 2026. doi: 10.1287/orsc.2025.21838. URL <https://pubsonline.informs.org/doi/10.1287/orsc.2025.21838>. Originally Harvard Business School Working Paper 24-013 (2023).
- Pablo Dorta-González, Alexis Jorge López-Puig, María Isabel Dorta-González, and Sara M. González-Betancor. Generative artificial intelligence usage by researchers at work: Effects of gender, career stage, type of workplace, and perceived barriers. *Telematics and Informatics*, 94: 102187, 2024. doi: 10.1016/j.tele.2024.102187. URL <https://doi.org/10.1016/j.tele.2024.102187>.
- Anil R. Doshi and Oliver P. Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28):eadn5290, 2024. doi: 10.1126/sciadv.adn5290. URL <https://www.science.org/doi/10.1126/sciadv.adn5290>.
- Elsevier. Insights 2024: Attitudes toward AI. Technical report, Elsevier, 2024. URL <https://www.elsevier.com/insights/attitudes-toward-ai>. Survey of ~3,000 researchers and clinicians across 123 countries.
- Jian Gao and Dashun Wang. Quantifying the use and potential benefits of artificial intelligence in scientific research. *Nature Human Behaviour*, 8:2281–2292, 2024. doi: 10.1038/s41562-024-02020-5. URL <https://www.nature.com/articles/s41562-024-02020-5>.
- Shanghai Gao, Ada Fang, Yepeng Huang, Valentina Giunchiglia, Ayush Noori, Jonathan Richard Schwarz, Yasha Ektefaie, Jovana Kondic, and Marinka Zitnik. Empowering biomedical discovery with AI agents. *Cell*, 187(22):6125–6151, 2024. doi: 10.1016/j.cell.2024.09.022. URL [https://www.cell.com/cell/fulltext/S0092-8674\(24\)01070-5](https://www.cell.com/cell/fulltext/S0092-8674(24)01070-5).
- Ali Essam Ghareeb, Michael Zhu, Jon M. Laurent, Samuel G. Rodrigues, et al. Robin: A multi-agent system for automating scientific discovery. *arXiv preprint arXiv:2505.13400*, 2025. doi: 10.48550/arXiv.2505.13400. URL <https://arxiv.org/abs/2505.13400>.
- Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, Anil Palepu, Petar Sirkovic, et al. Towards an AI co-scientist. *arXiv preprint arXiv:2502.18864*, 2025. doi: 10.48550/arXiv.2502.18864. URL <https://arxiv.org/abs/2502.18864>.

- Andrew Gray. ChatGPT “contamination”: estimating the prevalence of LLMs in the scholarly literature, 2024. URL <https://arxiv.org/abs/2403.16887>.
- Jutta Haider, Kristofer Rolf Söderström, Björn Ekström, and Malte Rödl. GPT-fabricated scientific papers on Google Scholar: Key features, spread, and implications for preempting evidence manipulation. *Harvard Kennedy School Misinformation Review*, 2024. doi: 10.37016/mr-2020-156. URL <https://misinforeview.hks.harvard.edu/article/gpt-fabricated-scientific-papers-on-google-scholar-key-features-spread-and-implications-for-preempting-evidence-manipulation/>.
- Qianyu Hao, Fengli Xu, Yong Li, and James Evans. Artificial intelligence tools expand scientists’ impact but contract science’s focus. *Nature*, 2025. doi: 10.1038/s41586-025-09922-y. URL <https://www.nature.com/articles/s41586-025-09922-y>.
- Sanchaita Hazra, Doeun Lee, Bodhisattwa Prasad Majumder, and Sachin Kumar. Accepted with minor revisions: Value of AI-assisted scientific writing. In *Proceedings of the 2026 ACM Conference on Intelligent User Interfaces (IUI)*, 2026. URL <https://arxiv.org/abs/2511.12529>.
- International Committee of Medical Journal Editors. Recommendations for the conduct, reporting, editing, and publication of scholarly work in medical journals: Defining the role of authors and contributors. ICMJE, 2023. URL <https://www.icmje.org/recommendations/browse/roles-and-responsibilities/defining-the-role-of-authors-and-contributors.html>. AI-assisted technology section added May 2023.
- Jacqueline A. Jansen, Artür Manukyan, Nour Al Khoury, and Altuna Akalin. Leveraging large language models for data analysis automation. *PLOS ONE*, 20(2):e0317084, 2025. doi: 10.1371/journal.pone.0317084. URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0317084>.
- JetBrains. The state of developer ecosystem 2024. Industry survey report, JetBrains, December 2024. URL <https://www.jetbrains.com/lp/devecosystem-2024/>. Survey conducted May–June 2024; 23,262 respondents after cleaning.
- Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2310.06770>.
- Minseok Jung, Aurora Zhang, May Fung, Junho Lee, and Paul Pu Liang. Quantitative insights into large language model usage and trust in academia: An empirical study, 2025. URL <https://arxiv.org/abs/2409.09186>.
- Shivani Kapania, Ruiyi Wang, Toby Jia-Jun Li, Tianshi Li, and Hong Shen. “i’m categorizing LLM as a productivity tool”: Examining ethics of LLM use in HCI research practices. 2024. URL <https://arxiv.org/abs/2403.19876>.
- Shivani Kapania, William Agnew, Motahhare Eslami, Hoda Heidari, and Sarah Fox. ‘simulacrum of stories’: Examining large language models as qualitative research participants. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025. doi: 10.1145/3706598.3713220. URL <https://dl.acm.org/doi/full/10.1145/3706598.3713220>.

- Dmitry Kobak, Rita González-Márquez, Emőke-Ágnes Horvát, and Jan Lause. Delving into LLM-assisted writing in biomedical publications through excess vocabulary. *Science Advances*, 11(27): eadt3813, 2025. doi: 10.1126/sciadv.adt3813. URL <https://www.science.org/doi/10.1126/sciadv.adt3813>.
- Nataliya Kosmyna, Eugene Hauptmann, Ye Tong Yuan, Jessica Situ, Xian-Hao Liao, Ashly Vivian Beresnitzky, Iris Braunstein, and Pattie Maes. Your brain on ChatGPT: Accumulation of cognitive debt when using an AI assistant for essay writing task, 2025. URL <https://arxiv.org/abs/2506.08872>.
- Diana Kwon. Is it OK for AI to write science papers? Nature survey shows researchers are split. *Nature*, 641(8063):574–578, 2025. doi: 10.1038/d41586-025-01463-8. URL <https://www.nature.com/articles/d41586-025-01463-8>.
- Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou. GPT detectors are biased against non-native English writers. *Patterns*, 4(7):100779, 2023. doi: 10.1016/j.patter.2023.100779. URL <https://doi.org/10.1016/j.patter.2023.100779>.
- Weixin Liang, Zachary Izzo, Yaohui Zhang, Haley Lepp, Hancheng Cao, Xuandong Zhao, Lingjiao Chen, Haotian Ye, Sheng Liu, Zhi Huang, Daniel A. McFarland, and James Y. Zou. Monitoring AI-modified content at scale: A case study on the impact of ChatGPT on AI conference peer reviews. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2403.07183>.
- Weixin Liang, Yaohui Zhang, Zhengxuan Wu, Haley Lepp, Wenlong Ji, Xuandong Zhao, Hancheng Cao, Sheng Liu, Siyu He, Zhi Huang, Diyi Yang, Christopher Potts, Christopher D. Manning, and James Zou. Quantifying large language model usage in scientific papers. *Nature Human Behaviour*, 9:2599–2609, 2025. doi: 10.1038/s41562-025-02273-8. URL <https://www.nature.com/articles/s41562-025-02273-8>.
- Bowen Lou, Tian Lu, T. S. Raghu, and Yingjie Zhang. Unraveling human–AI teaming: A review and outlook, 2025. URL <https://arxiv.org/abs/2504.05755>.
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024. doi: 10.48550/arXiv.2408.06292. URL <https://arxiv.org/abs/2408.06292>.
- Sasha Luccioni, Yacine Jernite, and Emma Strubell. Power hungry processing: Watts driving the cost of AI deployment? In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 85–99, 2024. doi: 10.1145/3630106.3658542. URL <https://doi.org/10.1145/3630106.3658542>.
- Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5):525–535, 2024. doi: 10.1038/s42256-024-00832-8. URL <https://www.nature.com/articles/s42256-024-00832-8>.
- Lisa Messeri and M. J. Crockett. Artificial intelligence and illusions of understanding in scientific research. *Nature*, 627:49–58, 2024. doi: 10.1038/s41586-024-07146-0. URL <https://www.nature.com/articles/s41586-024-07146-0>.

- Meredith Ringel Morris, Jascha Sohl-Dickstein, Noah Fiedel, Tris Warkentin, Allan Dafoe, Aleksandra Faust, Clement Farabet, and Shane Legg. Position: Levels of AGI for operationalizing progress on the path to AGI. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2311.02462>.
- Miryam Naddaf. How are researchers using AI? survey reveals pros and cons for science. *Nature*, 2025a. doi: 10.1038/d41586-025-00343-5. URL <https://www.nature.com/articles/d41586-025-00343-5>. News coverage of Wiley’s ExplanAItions study.
- Miryam Naddaf. More than half of researchers now use AI for peer review — often against guidance. *Nature*, 2025b. doi: 10.1038/d41586-025-04066-5. URL <https://www.nature.com/articles/d41586-025-04066-5>. News coverage of a Frontiers survey of ~1,600 academics.
- Siddharth Narayanan, James D. Braza, Ryan-Rhys Griffiths, Manvitha Ponnampati, Albert Bou, Jon M. Laurent, et al. Aviary: training language agents on challenging scientific tasks. *arXiv preprint arXiv:2412.21154*, 2024. doi: 10.48550/arXiv.2412.21154. URL <https://arxiv.org/abs/2412.21154>.
- National Institutes of Health. The use of generative artificial intelligence technologies is prohibited for the NIH peer review process. NIH Guide Notice NOT-OD-23-149, June 2023. URL <https://grants.nih.gov/grants/guide/notice-files/NOT-OD-23-149.html>. Released 23 June 2023.
- Nature Editorial. Tools such as ChatGPT threaten transparent science; here are our ground rules for their use. *Nature*, 613:612, 2023. doi: 10.1038/d41586-023-00191-1. URL <https://www.nature.com/articles/d41586-023-00191-1>.
- Mohamed Nejjar, Luca Zacharias, Fabian Stiehle, and Ingo Weber. LLMs for science: Usage for code generation and data analysis. *Journal of Software: Evolution and Process*, 37(1):e2723, 2025. doi: 10.1002/smr.2723. URL <https://onlinelibrary.wiley.com/doi/full/10.1002/smr.2723>.
- Brenda Nogueira, Werner Geyer, Andrew Anderson, Toby Jia-Jun Li, Dongwhi Kim, Nuno Moniz, and Nitesh V. Chawla. From verification burden to trusted collaboration: Design goals for LLM-assisted literature reviews, 2025. URL <https://arxiv.org/abs/2512.11661>.
- Linda Nordling. How ChatGPT is transforming the postdoc experience. *Nature*, 622(7983):655–657, 2023. doi: 10.1038/d41586-023-03235-8. URL <https://www.nature.com/articles/d41586-023-03235-8>.
- Linda Nordling. How money, politics and technology are redefining the PhD experience in 2025. *Nature*, 646(8083), 2025. doi: 10.1038/d41586-025-03149-7. URL <https://www.nature.com/articles/d41586-025-03149-7>.
- Alexander Novikov, Ngân Vũ, Marvin Eisenberger, et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery. *arXiv preprint arXiv:2506.13131*, 2025. doi: 10.48550/arXiv.2506.13131. URL <https://arxiv.org/abs/2506.13131>.
- Gabrielle O’Brien, Alexis Parker, Nasir U. Eisty, and Jeffrey C. Carver. A survey of generative AI adoption and perceived productivity among scientists who program, 2025. URL <https://arxiv.org/abs/2512.19644>.
- Brian Owens. How Nature readers are using ChatGPT. *Nature*, 615(7950):20, 2023. doi: 10.1038/d41586-023-00500-8. URL <https://www.nature.com/articles/d41586-023-00500-8>.

- Oxford University Press. Researchers and AI: Survey findings. Technical report, Oxford University Press, 2024. URL <https://academic.oup.com/pages/ai-survey-findings>. Survey of 2,345 researchers across disciplines and career stages.
- Alexis Palmer, Noah A. Smith, and Arthur Spirling. Using proprietary language models in academic research requires explicit justification. *Nature Computational Science*, 4(1):2–3, 2024. doi: 10.1038/s43588-023-00585-1. URL <https://www.nature.com/articles/s43588-023-00585-1>.
- Sida Peng, Eirini Kalliamvakou, Peter Cihon, and Mert Demirer. The impact of AI on developer productivity: Evidence from GitHub Copilot. *arXiv preprint arXiv:2302.06590*, 2023. doi: 10.48550/arXiv.2302.06590. URL <https://arxiv.org/abs/2302.06590>.
- Yifan Qian, Zhe Wen, Alexander C. Furnas, Yue Bai, Erzhao Shao, and Dashun Wang. The rise of large language models and the direction and impact of US federal research funding, 2026. URL <https://arxiv.org/abs/2601.15485>.
- Steven Randazzo, Hila Lifshitz-Assaf, Katherine C. Kellogg, Fabrizio Dell’Acqua, Ethan R. Mollick, François Cadelon, and Karim R. Lakhani. Cyborgs, centaurs and self-automators: The three modes of human–genai knowledge work and their implications for skilling and the future of expertise. Working Paper 26-036, Harvard Business School, 2025. URL <https://www.hbs.edu/faculty/Pages/item.aspx?num=68273>.
- Giuseppe Russo, Manoel Horta Ribeiro, Tim R. Davidson, Veniamin Veselovsky, and Robert West. The AI review lottery: Widespread AI-assisted peer reviews boost paper scores and acceptance rates. *Proceedings of the ACM on Human-Computer Interaction (CSCW)*, 9(7), 2025. doi: 10.1145/3757667. URL <https://dl.acm.org/doi/10.1145/3757667>.
- Adrian Ryser, Florian Allwein, and Tim Schlippe. Calibrated trust in dealing with LLM hallucinations: A qualitative study. In *Proceedings of the 3rd International Conference on Foundation and Large Language Models (FLLM)*, 2025. URL <https://arxiv.org/abs/2512.09088>.
- Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zicheng Liu, and Emad Barsoum. Agent laboratory: Using LLM agents as research assistants. *arXiv preprint arXiv:2501.04227*, 2025. doi: 10.48550/arXiv.2501.04227. URL <https://arxiv.org/abs/2501.04227>.
- Hope Schroeder, Marianne Aubin Le Quéré, Casey Randazzo, David Mimno, and Sarita Schoenebeck. Large language models in qualitative research: Uses, tensions, and intentions. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025. doi: 10.1145/3706598.3713120. URL <https://arxiv.org/abs/2410.07362>.
- Erzhao Shao, Yifan Wang, Yifan Qian, Zhenyu Pan, Han Liu, and Dashun Wang. SciSciGPT: advancing human–AI collaboration in the science of science. *Nature Computational Science*, 6: 301–315, 2025. doi: 10.1038/s43588-025-00906-6. URL <https://www.nature.com/articles/s43588-025-00906-6>.
- Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. Can LLMs generate novel research ideas? a large-scale human study with 100+ NLP researchers. *arXiv preprint arXiv:2409.04109*, 2024. doi: 10.48550/arXiv.2409.04109. URL <https://arxiv.org/abs/2409.04109>.
- Chenglei Si, Tatsunori Hashimoto, and Diyi Yang. The ideation-execution gap: Execution outcomes of LLM-generated versus human research ideas. *arXiv preprint arXiv:2506.20803*, 2025. doi: 10.48550/arXiv.2506.20803. URL <https://arxiv.org/abs/2506.20803>.

- Michael D. Skarlinski, Sam Cox, Jon M. Laurent, James D. Braza, Michaela Hinks, Michael J. Hammerling, Manvitha Ponnampati, Samuel G. Rodrigues, and Andrew D. White. Language agents achieve superhuman synthesis of scientific knowledge. *arXiv preprint arXiv:2409.13740*, 2024. doi: 10.48550/arXiv.2409.13740. URL <https://arxiv.org/abs/2409.13740>.
- Stack Overflow. AI — 2024 Stack Overflow developer survey. Developer survey, Stack Overflow, 2024. URL <https://survey.stackoverflow.co/2024/ai>. Section: AI. 65,437 respondents worldwide.
- Stack Overflow. AI — 2025 Stack Overflow developer survey. Developer survey, Stack Overflow, 2025. URL <https://survey.stackoverflow.co/2025/ai>. Section: AI.
- Viktor Stray, Elias Goldmann Brandtzæg, Viggo Tellefsen Wivestad, Astri Barbala, and Nils Brede Moe. Developer productivity with and without GitHub Copilot: A longitudinal mixed-methods case study. In *Proceedings of the 59th Hawaii International Conference on System Sciences (HICSS)*, 2026. URL <https://arxiv.org/abs/2509.20353>.
- Kyle Swanson, Wesley Wu, Nash L. Bulaong, John E. Pak, and James Zou. The virtual lab of AI agents designs new SARS-CoV-2 nanobodies. *Nature*, 646:716–723, 2025. doi: 10.1038/s41586-025-09442-9. URL <https://www.nature.com/articles/s41586-025-09442-9>.
- Nihar Thakkar, Mert Yuksekgonul, Jonah Silberg, et al. A large-scale randomized study of large language model feedback in peer review. *Nature Machine Intelligence*, 8:326–336, 2026. doi: 10.1038/s42256-026-01188-x. URL <https://www.nature.com/articles/s42256-026-01188-x>.
- H. Holden Thorp. ChatGPT is fun, but not an author. *Science*, 379(6630):313, 2023. doi: 10.1126/science.adg7879. URL <https://www.science.org/doi/10.1126/science.adg7879>.
- U.S. National Science Foundation. Notice to the research community: Use of generative artificial intelligence technology in the NSF merit review process. NSF, December 2023. URL <https://www.nsf.gov/news/notice-to-the-research-community-on-ai>. Issued 14 December 2023.
- Richard Van Noorden. More than 10,000 research papers were retracted in 2023 — a new record. *Nature*, 624:479–481, 2023. doi: 10.1038/d41586-023-03974-8. URL <https://www.nature.com/articles/d41586-023-03974-8>.
- Richard Van Noorden and Jeffrey M. Perkel. AI and science: what 1,600 researchers think. *Nature*, 621:672–675, 2023. doi: 10.1038/d41586-023-02980-0. URL <https://www.nature.com/articles/d41586-023-02980-0>.
- Kelly B. Wagman, Matthew T. Dearing, and Marshini Chetty. Generative AI uses and risks for knowledge workers in a science organization. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, 2025. doi: 10.1145/3706598.3713827. URL <https://dl.acm.org/doi/10.1145/3706598.3713827>.
- William H. Walters and Esther Isabelle Wilder. Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13:14045, 2023. doi: 10.1038/s41598-023-41032-5. URL <https://www.nature.com/articles/s41598-023-41032-5>.
- Jiaqi Wei, Yuejin Yang, Xiang Zhang, et al. From AI for science to agentic science: A survey on autonomous scientific discovery. *arXiv preprint arXiv:2508.14111*, 2025. doi: 10.48550/arXiv.2508.14111. URL <https://arxiv.org/abs/2508.14111>.

- Wiley. Explanaitions: An AI study by Wiley. Technical report, John Wiley and Sons, 2025. URL <https://www.wiley.com/en-us/about-us/ai-resources/ai-study/>. Multi-wave survey; up to ~4,946 researchers across 70+ countries.
- Huimin Xu, Houjiang Liu, Yan Leng, and Ying Ding. Adapting to LLMs: How insiders and outsiders reshape scientific knowledge production, 2025. URL <https://arxiv.org/abs/2505.12666>.
- Yutaro Yamada, Robert Tjarko Lange, Cong Lu, Shengran Hu, Chris Lu, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist-v2: Workshop-level automated scientific discovery via agentic tree search. *arXiv preprint arXiv:2504.08066*, 2025. doi: 10.48550/arXiv.2504.08066. URL <https://arxiv.org/abs/2504.08066>.
- John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL <https://arxiv.org/abs/2405.15793>.